# The Lens Maker’s Equation

> SYNOPSIS
>
> A lens is two curved refracting surfaces back to back, and a ray crossing each one bends by Snell’s law — the law of Note 3, nothing new. Do that twice and out falls a single number, the focal length, written in terms of the two radii and the glass. This note builds that equation and then throws most of it away.

**BEFORE YOU START — PREDICT**

Two lenses are ground from the same glass to the same shape, but one is made **twice as thick through the middle**. Do they have the same focal length? Commit to an answer. Then read the last line of this note.

**PERFORMANCE EXPECTATIONS**

1. **Identify** $R$ on a lens surface as the radius of the circle that surface is a piece of.
2. **State** the lens maker’s equation and say what each of $f$, $R_1$, $R_2$, $n_1$, $n_2$ and $d$ stands for.
3. **Reduce** it to the thin-lens-in-air form, and say exactly which two assumptions you used.
4. **Compute** a focal length from two radii and a refractive index.

## 1. Lens maker’s equation

**A ray of light travelling through a curved surface will bend according to Snell’s law.** That is the entire physical input. Everything below is geometry applied to it twice — once going in, once coming out.

***R* = radius of curvature**, which describes the edge of the refracting surface.

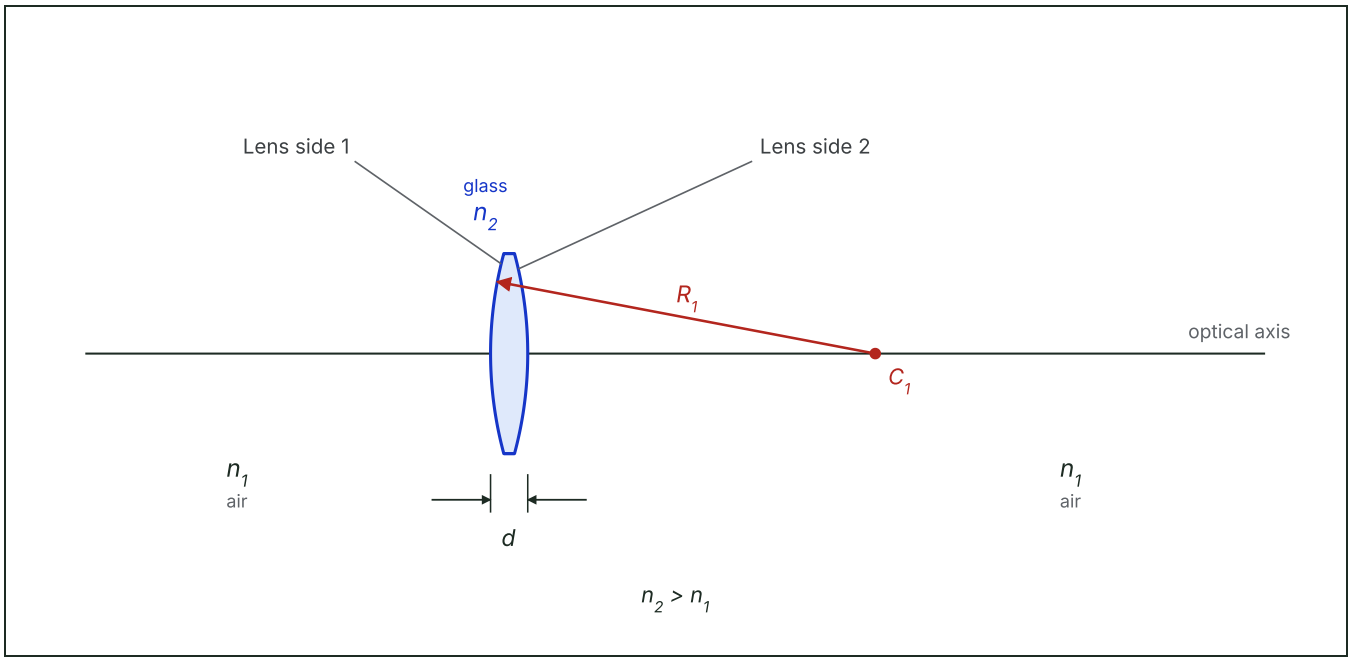



**Figure 1.** Every curved surface in this note is a piece of a circle, and $R$ is that circle's radius — drawn here from each real centre of curvature out to its own real arc. **Both arcs above have the same radius**; they get opposite signs in the equation only because they bulge in opposite directions.

**Your job on Figure 1.** Lens side 1 is constructed for you: $C_1$ is marked and $R_1$ drawn from it out to the arc. Do the same for side 2 — mark its centre of curvature and draw $R_2$. Which side of the lens does that centre fall on, and why does that give $R_2$ the opposite sign?

Carrying Snell's law across both surfaces and collecting the terms gives the focal length of the lens:

$$1/f = (n_2/n_1 - 1)\,[\,1/R_1 - 1/R_2 + (n_2 - 1)d\,/\,(n_2 R_1 R_2)\,]$$

- $f$ — focal length of the lens
- $R_1$ — radius of curvature of lens side 1
- $R_2$ — radius of curvature of lens side 2
- $n_1$ — refractive index of the medium in which the object sits
- $n_2$ — refractive index of the lens
- $d$ — thickness of the lens

## The form you will actually use

For a **thin lens in air**, two things are true at once: $d \ll R$, so the last term inside the bracket is negligible; and $n_1 = 1$, so $n_2/n_1$ is just the index of the glass, $n$. Both assumptions together collapse the equation to

$$1/f = (n - 1)\,[\,1/R_1 - 1/R_2\,]$$

**THE PUNCHLINE**

$$1/f = (n - 1)\,[\,1/R_1 - 1/R_2\,]$$

Back to the prediction: the thickness $d$ dropped out. Two lenses of the same glass and the same curvature have the **same focal length** however thick they are through the middle — as long as both are thin enough for $d \ll R$ to hold. Make one thick enough and the term comes back.

## Looking ahead

The lens maker's equation is about *grinding* a lens — glass and curvature in, focal length out. From **Note 6** onward the focal length is simply given to you, and the question becomes where the image lands.

2. Which **two** assumptions take the full equation to $1/f = (n - 1)[1/R_1 - 1/R_2]$?

◯ The lens is thin, and it sits in air. ◯ The lens is thin, and the object is far away. ◯ The two radii are equal, and the lens sits in air. ◯ The light is monochromatic, and the rays are close to the axis.

3. A thin lens in air has $n = 1.60$, $R_1 = +12$ cm and $R_2 = -12$ cm. Find $f$, in cm.

_____ cm

4. Two lenses are ground from the same glass to the same two radii, but lens B is twice as thick through the middle. Both are thin. How do their focal lengths compare?

◯ B's focal length is half of A's. ◯ B's focal length is twice A's. ◯ They are the same: $d$ has dropped out of the thin-lens form. ◯ It cannot be decided without knowing the lens diameter.

Check Clear

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